

Expt. No. : 4

Date :

BUILD A PROGRAM TO PERFORM ARITHMETIC OPERATIONS USING LAMBDA FUNCTION

AIM

To write a python program to perform arithmetic operations using lambda function.

SOFTWARE REQUIRED

Python 3.14

SYNTAX

lambda arguments: expression

THEORY

Lambda Functions in Python are anonymous functions, implying they don't have a name. The def keyword is needed to create a typical function in Python, as we already know. We can also use the lambda keyword in Python to define an unnamed function. Syntax: lambda arguments: expression This function accepts any count of inputs but only evaluates and returns one expression. That means it takes many inputs but returns only one output.

PROBLEM

Build a Python program to calculate the square root and power of a given number using lambda functions.

APPLICATION PROBLEM

A testing laboratory evaluates the efficiency of electrical machines by comparing the input and output power. Build a Python program using lambda functions to calculate the efficiency of a motor or generator.

ALGORITHM

STEP 1: Create lambda functions for each arithmetic operation: add, subtract, multiply, and divide.

STEP 2: Each lambda function should take two arguments (x and y) and return the result of the corresponding arithmetic operation.

STEP 3: Pass the lambda functions and operands to a function that performs the arithmetic operations.

STEP 4: The function should take the operation (lambda function) and operands (x and y) as arguments.

STEP 5: Execute the lambda function with the provided operands to compute the result.

STEP 6: Return the result of the arithmetic operation.

STEP 7: Terminate the program.

PROGRAM CODING

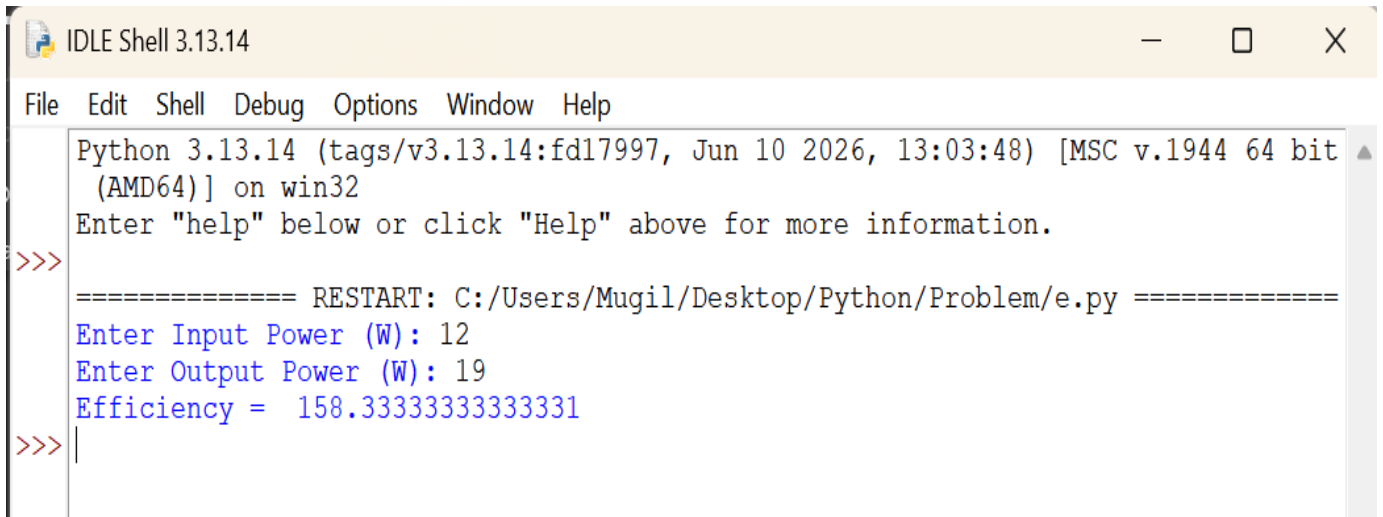
```
num = float(input("Enter a number to take square root: "))  
power = int(input("Enter the power value for the number: "))  
square_root = lambda x: x**0.5  
power_value = lambda x, y: x ** y  
print("Square Root =", square_root(num))  
print("Power =", power_value(num, power))
```

OUTCOMES OF PROGRAM CODING

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on
win32
Enter "help" below or click "Help" above for more information.
>>>
==== RESTART: C:\Users\Mugil\Desktop\Python\Experiment\Experiment4.py =====
Enter a number to take square root: 12
Enter the power value for the number: 12
Square Root = 3.4641016151377544
Power = 8916100448256.0
>>>
```

APPLICATION PROBLEM CODING

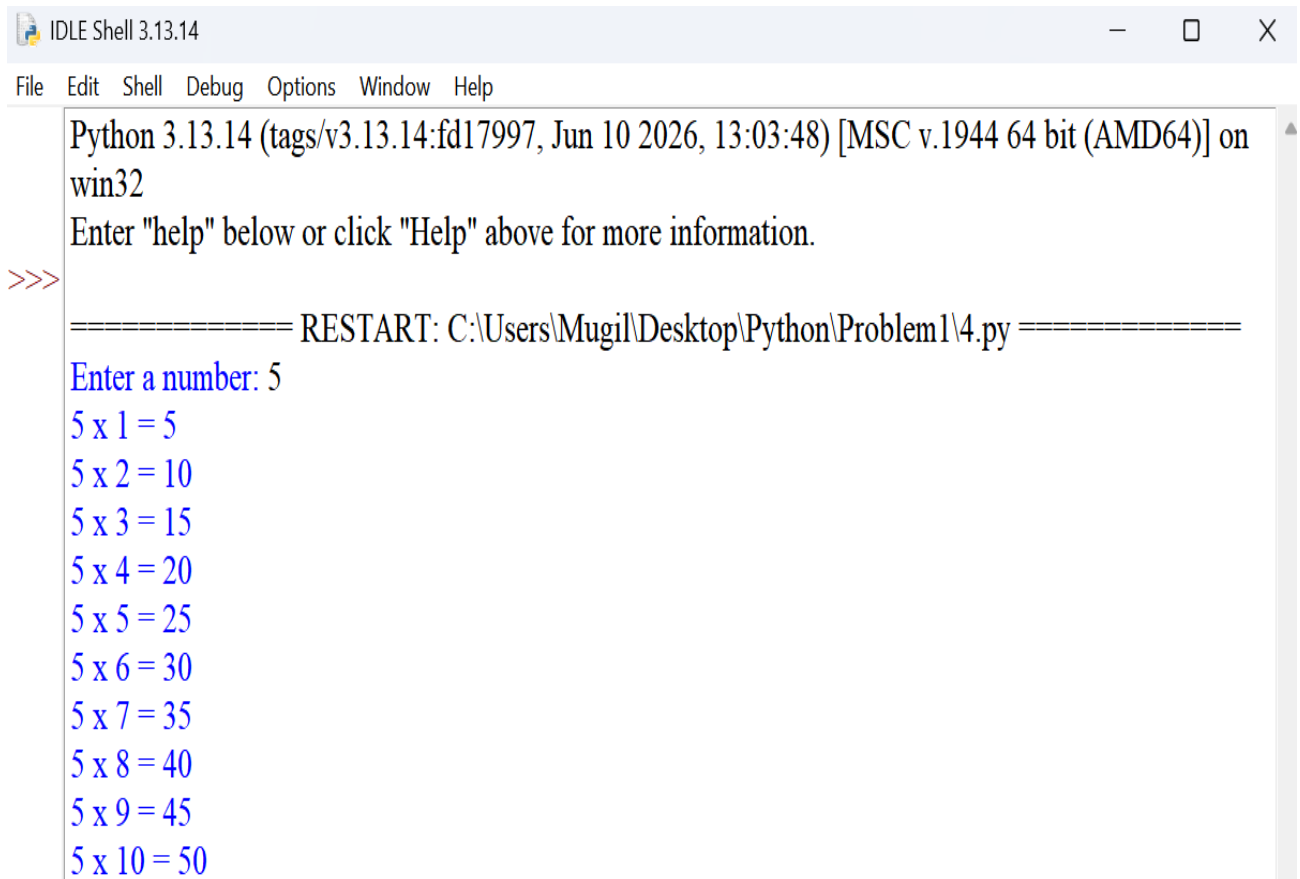
```
input_power = float(input("Enter Input Power (W): "))
output_power = float(input("Enter Output Power (W): "))
efficiency = lambda op, ip: (op / ip) * 100
print("Efficiency = ",efficiency(output_power,input_power))
```

OUTCOMES OF APPLICATION PROGRAM CODING

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Mugil/Desktop/Python/Problem/e.py =====
Enter Input Power (W): 12
Enter Output Power (W): 19
Efficiency = 158.33333333333331
>>> |
```

PROBLEM CODING

```
n = int(input("Enter a number: "))
multiply = lambda x, y: x * y
for i in range(1, 11):
    print(n, "x", i, "=", multiply(n, i))
```

OUTCOMES OF PROGRAM CODING

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=====RESTART: C:\Users\Mugil\Desktop\Python\Problem1\4.py=====
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

VIVA QUESTION

1. What are *args and *kwargs?

*args allows a function to accept any number of positional arguments as a tuple.

**kwargs allows a function to accept any number of keyword arguments as a dictionary.

2. Is Indentation Required in Python?

Yes, indentation is required in Python. It is used to define the structure of code blocks such as loops, conditionals, and functions.

3. What is Scope in Python?

Scope refers to the visibility and lifetime of variables in different parts of the program. It includes local, enclosing, global, and built-in scopes.

4. What is docstring in Python?

A docstring is a string literal used to document a function, class, or module. It is typically placed right after the definition of a function or class to provide information about its purpose.

5. What is a dynamically typed language?

A dynamically typed language is one where the type of a variable is determined at runtime, and variables can change types without needing explicit type declarations.

RESULT:

Thus, the arithmetic operation is performed using python lambda function and its application program was also executed and verified successfully.

INFERENCE:

The experiment demonstrates that Python lambda functions can efficiently perform arithmetic operations using concise and anonymous functions. The successful execution and verification of the program confirm the correctness of the implemented logic and highlight the effectiveness of lambda functions in simplifying mathematical computations and real-world engineering applications.

Assessment	Max. Marks	Marks Obtained
Conduction of Practical	15	
Report	5	
Viva	5	
Total Marks	25	
Signature of Lab In-Charge		